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SYSTEMS, METHODS, AND COMPUTER PROGRAM PRODUCT FOR CONTENT EXCHANGE SERVICES FOR PAYMENT NETWORKS

Abstract

Methods, systems, and computer program products are provided for secure content exchange. A secure content exchange platform includes a content exchanger and exchange gateway interface. The content exchanger stores a content item and generates a globally unique identifier (GUID) associated with the content item. The exchange gateway interface receives a registration request for the content item from a sending financial institution system, submits the content item to the content exchanger, receives the GUID associated with the content item from the content exchanger, returns the GUID to the sending financial institution system as a response to the registration request, receives a retrieval request from a recipient financial institution system, retrieves the GUID from the retrieval request to submit the GUID to the content exchanger, receives the content item from the content exchanger, and returns the content item to the recipient financial institution system as a response to the retrieval request.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority to and is a Continuation of U.S. patent application Ser. No. 18/185,510, filed Mar. 17, 2023, which application is incorporated herein by reference in its entirety.

BACKGROUND

Field

[0002] Example aspects of the present disclosure generally related to electronic transactions, and more particularly related to electronic transactions across payment networks with associated content items.

Related Art

[0003] The exchange of information between computing systems operated by different entities is an integral part of modern business transactions. In the context of payment networks, transaction-related information can include documentation related to the ultimate payment network credit or debit transaction communications between such computing systems, such as receipts or remittance documentation. The exchange of transaction-related information between, for example, payor systems and payee systems are predominately facilitated via email and other unsecure channels whereas the credit or debit communications must be kept secure. This poses significant challenges in terms of security and efficiency and can create a disconnect between the transaction-related information, such as invoices and remittance documentation and the associated credit or debit transaction. In turn, this disconnect not only creates inefficiency within receivables and payables processes, but also a poor user experience for all the involved parties.

[0004] The technical challenges associated with the exchange of transaction-related information between computing systems operated by different entities involve security, interoperability, data standardization, integration, scalability, and timing of transactions including timing of transaction-related information. Overall, the technical challenges associated with the exchange of transaction-related information between computing systems operated by different entities are significant as the inhibit efficient and secure processing of transactions.

SUMMARY

[0005] The foregoing problems with security and connectivity in the current state of payment systems, as well as other limitations are overcome by systems, methods and computer program products for a secure content exchange platform as disclosed herein.

[0006] Example aspects of the present disclosure relate to a system for secure content exchange. The system includes a content exchanger configured to store a content item and generate a globally unique identifier (GUID) associated with the content item. The system further includes an exchange gateway interface in communication with the content exchanger and configured to: receive a registration request for the content item from a sending financial institution system; submit the content item to the content exchanger; receive the GUID associated with the content item from the content exchanger; return the GUID to the sending financial institution system as a response to the registration request; receive a retrieval request from a recipient financial institution system; retrieve the GUID from the retrieval request; submit the GUID to the content exchanger; receive the content item from the content exchanger; and return the content item to the recipient

financial institution system as a response to the retrieval request.

[0007] In one embodiment, the sending financial institution system transmits the GUID in a message. In some implementations, the message is transmitted using a payment network. In another embodiment, the payment network is a real time payment network.

[0008] In another embodiment, the exchange gateway interface is further configured to register the content item by validating a first credential associated with the sending financial institution system. In some implementations, the content item originates from a content provider system and the exchange gateway interface is further configured to validate a second credential associated with the content provider system. In other aspects, the exchange gateway interface is further configured to retrieve the content item by validating a third credential associated with the recipient financial institution system. In some further aspects, the message is directed to a content recipient associated with the recipient financial institution system, wherein the exchange gateway interface is further configured to validate a fourth credential associated with the content recipient system.

[0009] In another embodiment, the system further includes at least one management portal. In some implementations, the at least one management portal in communication with one or more of a registration application programming interface (API) or a retrieval API. In other aspects, the system further includes a first management portal associated with the sending financial institution system and a second management portal associated with the recipient financial institution system. In another embodiment, the content item is one of a document, a data file, or an image.

[0010] Further example aspects of the present disclosure relate to a method for securely exchanging content items via a secure content exchange server. The method includes steps of receiving, from a sending financial institution system, a registration request for a content item; submitting, to a content exchanger, the content item; receiving, from the content exchanger, a globally unique identifier (GUID) associated with the content item; returning the GUID to the sending financial institution system; receiving, from a recipient financial institution system, a retrieval request including the GUID; retrieving, from the message, the GUID; submitting, to the content exchanger, the GUID; receiving, from the content exchanger, the content item; and returning, to the recipient financial institution system, the content item.

[0011] In an embodiment, the message is transmitted from the sending financial institution system to the recipient financial institution system via a payment network. In some implementations, the payment network is a real time payment network. In other aspects, the method further includes steps of receiving, from the sending financial institution system, one or more credentials; and verifying the one or more credentials.

[0012] In some implementations, the one or more credentials comprise credentials for a content provider system associated with the content item. In further aspects, the one or more credentials comprise credentials for the sending financial institution system. In other embodiments, the sending financial institution system submits the registration request via a registration API.

[0013] A variety of additional inventive aspects will be set forth in the description that follows. The inventive aspects can relate to individual features and to combinations of features. It is to be understood that both the forgoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the embodiments disclosed herein are based.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The features and advantages of the example embodiments of the invention presented herein will become more apparent from the detailed description set forth below when taken in conjunction with the following drawings.

[0015] FIG. 1 depicts a system-flow diagram illustrating an architecture for exchanging payment messages with an associated content item, in accordance with an example embodiment.

[0016] FIG. 2 depicts a system-flow diagram illustrating a flow path of a content item through the content exchange system of FIG. 1, in accordance with an embodiment.

[0017] FIG. 3 depicts a system-flow diagram illustrating a flow path of a global unique identifier (GUID) through the content exchange system of FIG. 1, in accordance with an example embodiment.

[0018] FIG. 4 illustrates a flow chart of a method for registration of a content item, in accordance with an example embodiment.

[0019] FIG. 5 illustrates a flow chart of a method for retrieval of a content item, in accordance with an example embodiment.

[0020] FIG. 6 illustrates a flow chart of a method for updating a registered content item, in accordance with an example embodiment.

[0021] FIG. 7 illustrates a communication sequence between a financial institution infrastructure and a secure content exchange platform, in accordance with an example embodiment.

[0022] FIG. 8 illustrates a block diagram of a virtual or physical computing system, in accordance with an example embodiment.

DETAILED DESCRIPTION

[0023] In the following detailed description, references are made to the accompanying drawings that form a part hereof, and in which are shown by way of illustrations specific embodiments or examples. These aspects may be combined, other aspects may be utilized, and structural changes made without departing from the present disclosure. Embodiments may be practiced as methods, systems, or devices. Accordingly, embodiments may take the form of a hardware implementation, an entirely software implementation, or an implementation combining software and hardware aspects. The following detailed description is therefore not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.

[0024] Generally, example aspects of the embodiments described herein provide systems, methods, and computer program products for securely exchanging content items associated with payment messages. More specifically, example aspects of the embodiments described herein integrate a secure content exchange platform for registration and retrieval of electronic documents and other content items of referential material to be transmitted for display from one party to another a part of a payment transaction. Examples of such referential materials include documents itemizing or describing what payment due is for or what accounts are to be reconciled by a payment. Example formats which content items may be prepared or received in include Extensible Markup Language (XML), SWIFT, X12 820, and others. Financial institutions communicate with the platform through cloud-based interfaces, such as application programming interfaces (APIs), on their client's behalf, supporting seamless integration with existing payment systems.

[0025] Example aspects of the present disclosure relate to a system for secure content exchange. The system includes a secure content exchange platform including a content exchanger configured to store a content item (e.g., a document, a data file, or an image or other media file) and generate a globally unique identifier (GUID) associated with the content item. The secure content exchange platform further includes an exchange gateway interface in communication with the content exchanger and configured to receive a registration request for the content item from a sending financial institution system and submit the content item to the content exchanger. The exchange gateway interface receives the GUID associated with the content item from the content exchanger and returns the GUID to the sending financial institution system as a response to the registration request. The exchange gateway interface is also configured to receive a retrieval request from a recipient financial institution system and retrieve the GUID from the retrieval request to submit the GUID to the content exchanger. The exchange gateway interface receives the content item from the content exchanger and returns the content item to the recipient financial institution system as a

response to the retrieval request.

[0026] Implementations of the document exchange platform disclosed herein overcome limitations in existing payment network transactions. The existing capabilities of banks and providers can be extended through use of a common infrastructure and the secure exchange of information between a payor and payee can be supported. Advantages for each participant in payment network transactions are provided by aspects of the present disclosure.

[0027] A payee's bank or other financial institution may experience an increase in the value of both retail and commercial products by leveraging a common set of capabilities. By addressing limitations of the present information exchange environment, the payee bank can support a broader market and drive additional volume of client support. A payor's bank or other financial institution may experience similar benefits.

[0028] A payee company or individual may themselves experience benefits such as reduced costs associated with invoicing and billing, and streamlining data exchange by direct association of supporting documents with payment request messages. Risk is also reduced by using the secure payment network channel to exchange all payment associated information with payors. Additionally, the presentation of the invoice or bill may be improved over the current capabilities of payment network exchanges. A payor company or individual may experience benefits such as reduced time and effort to receive, review, process, and pay invoices and bills due to streamlining of data exchange with a payee. Risk reduction and improvements in presentation are also advantageous to the payee.

[0029] Aspects of the present disclosure provide improved straight-through processing for payors and payees. Aspects of the present disclosure may contribute to the adoption and growth of electronic payment mechanisms through attributes such as enabling the seamless exchange of transaction-related artifacts, such as bills and invoices, between payor and payee; leveraging application programming interfaces (APIs) to support various capabilities within the payment and content exchange ecosystem (e.g., registration, retrieval, etc.); supporting an enhanced customer experience via payment networks; establishing connectivity; and performing certification and a range of other administrative tasks.

[0030] FIG. 1 depicts a system-flow diagram illustrating a content exchange system **100** for exchanging payment messages with secure associated content, in accordance with an example embodiment. In some implementations, the content exchange system **100** includes a sender system **102**, a sender financial institution system **104**, a payment network **106**, a recipient financial institution system **108**, a recipient system **110**, a registration interface **112**, a retrieval interface **114**, and a secure content exchange platform **116**. The secure content exchange platform **116** includes an exchange gateway interface **118**, a content exchanger **120**, and a content database **122**.

[0031] Sender system **102** operates to communicate with sender financial institution system **104**. In some use cases, sender system **102** is operated by a business or individual that is submitting content to be relayed to another system, such as recipient system **110**, in association with a payment message. In some embodiments, the payment message is a request for payment (RfP) message or a payment transfer (credit) message. In one example, the sender system **102** is associated with a supplier. The supplier submits a content item, such as an invoice document, to be transmitted with an RfP message to a recipient system **110** such as one operated by a buyer or other recipient entity.

[0032] Sender system **102** interacts, in some embodiments, with a portal associated with sender financial institution system **104**, to prepare a payment message with an associated content item. In some embodiments, the portal is a web-based portal or application, or a mobile application. In turn, when a content item is submitted to be registered, business rules are applied to validate the content. Rule may apply to the contents file size, file type, etc. In some implementations, rules are integrated with the portal.

[0033] Sender financial institution system **104** operates to communicate with sender system **102**, to receive a content item, and with secure content exchange platform **116** to register the content item.

In some use cases, sender financial institution system **104** is operated by business or other organization that provides financial services. Sender financial institution system **104** supports sender system **102** with the sending of the content item and the payment message. Sender financial institution system **104** registers a content item provided by sender system **102** with secure content exchange platform **116** using registration interface **112**. When a content item is registered, sender financial institution system **104** receives a globally unique identifier (GUID) via registration interface **112**. The GUID identifies the registered content item to secure content exchange platform **116**.

[0034] Sender financial institution system **104** prepares the payment message including the GUID and submits the payment message to payment network **106**. Continuing the example from above, sender financial institution system **104** receives the RfP payment message request and the invoice document to include from sender system **102**, who is a supplier. Sender financial institution system **104** registers the invoice document with secure content exchange platform **116** via registration interface **112** and receives a GUID associated with the invoice document. Sender financial institution system **104** prepares the RfP message including the GUID, and transmits the RfP message to recipient financial institution system **108** via payment network **106**.

[0035] Sender financial institution system **104** may communicate with registration interface **112** through a management portal. In some examples, the management portal is configured and maintained by sender financial institution system **104**. In some embodiments, sender financial institution system **104** also communicates with retrieval interface **114** and other interfaces and services according to the configuration of the management portal and the requests received from financial institution clients such as sender system **102** and recipient system **110**.

[0036] Payment network **106** is configured to relay a payment message from sender financial institution system **104** to recipient financial institution system **108**.

[0037] One such payment network **106** is referred to as a real-time payment network. Real-time payments, as used herein, are payments that are initiated, cleared and settled nearly instantaneously (e.g., within seconds) through the utilization of a payment network. A real-time payment network (also referred to as a real-time payments platform) generally refers to a network through which real-time payments are made. The RTP® network is a real-time payment network in the United States provided by The Clearing House. The RTP® network is accessible to financial institutions that hold DDAs. In the art, a real-time payment message is referred to as an RTP message. RTP in this context should not to be confused with the registered trademark RTP®.

[0038] Another type of payment network is the automated clearinghouse (ACH) system, referred to herein simply as the ACH network, is a network through which depository institutions send each other batches of electronic credit and debit transfers. The direct deposit of payroll, social security benefits, and tax refunds are typical examples of ACH credit transfers. The direct debiting of mortgages and utility bills are typical examples of ACH debit transfers. While the ACH network was originally used to process mostly recurring payments, the network is today being used extensively to process one-time debit transfers, such as converted check payments and payments made over the telephone and internet.

[0039] The Electronic Payment network (EPN) is an ACH system operated by The Clearing House that operates as a computerized electronic funds transfer system for processing both individual consumer and commercial financial transactions. The EPN and the Federal Reserve System are two national ACH platforms capable of performing ACH transactions.

[0040] As an ACH operator, the Reserve Banks (i.e., the regional banks of the Federal Reserve System) receive files of ACH payments from originating depository financial institutions, edit and sort the payments, deliver the payments to receiving depository financial institutions, and settle the payments by crediting and debiting the depository financial institutions' settlement accounts. The Federal Reserve System and EPN rely on each other to process interoperator ACH payments, payments in which the originating depository financial institution (ODFI) and the receiving

depository financial institution (RDFI) are served by different operators. These interoperator payments are typically settled by the so-called Reserve Bank Systems.

[0041] From a technology perspective, the RTP® network is significantly different from the ACH network. RTP® network payments clear and settle individually in real-time with immediate finality. Same day ACH payments are cleared in batches and finally settle after the payments clear. The RTP® network and ACH network are also built upon different infrastructures having their own protocols. A transaction that cannot be processed in real-time by accessing a demand deposit account over the RTP® network, for example, might be able to be processed over the ACH network as an ACH debit transaction.

[0042] One technical limitation of the existing payment networks such as the RTP® network and the ACH network is that they do not seamlessly communicate with each other. Moreover, the same controls that can be performed on the RTP® network cannot be performed on the ACH network. This is because the ACH network does not have immediate access to the same amount of information as the RTP® network. Further, tokens implemented in connection with these payment networks are not currently shared across all the connected entity systems. As such, integrating merchant payment systems, the RTP® network, and the ACH network as one ecosystem, much less doing so in a secure, controlled manner, is not presently possible.

[0043] In some embodiments, the payment network **106** is an RTP® network. In some embodiments, payment network **106** is an ACH network.

[0044] Payment network **106** routes (e.g., receive and transmit) electronic transactions and various types of messages between financial institutions, such as sender financial institution system **104** and recipient financial institution system **108**. Payment network **106** includes one or more computers and/or servers which are configured to handle electronic financial transactions. Payment network **106** includes one or more databases. Although represented as a single network, in some embodiments there are more than one such network. In some embodiments, payment network is a real-time payment network.

[0045] Recipient financial institution system **108** operates to communicate with secure content exchange platform **116**, to retrieve a content item, and with recipient system **110**, to present the content item. In some use cases, recipient financial institution system **108** is operated by business or other organization that provides financial services. Recipient financial institution system **108** supports an individual or business, such as recipient system **110**, with receiving the content item and the payment message. Recipient financial institution system **108** retrieves content items from secure content exchange platform **116** via retrieval interface **114** and presents the content item to recipient system **110**.

[0046] To retrieve a content item, recipient financial institution system **108** receives a payment message via payment network **106** and provides the payment message to recipient system **110**. If the payment message has a GUID included, the recipient system **110** may submit a retrieval request to recipient financial institution system **108**. Recipient financial institution system **108** receives the retrieval request from recipient system **110**, extracts the GUID from the payment message, and submits the retrieval request, including the GUID, to retrieval interface **114**. Recipient financial institution system **108** receives the content item from secure content exchange platform **116** via retrieval interface **114**, and presents the content item to recipient system **110**.

[0047] In some implementations, recipient financial institution system **108** communicates with retrieval interface **114** through a management portal. The management portal is configured and maintained by recipient financial institution system **108**, in some embodiments. In some embodiments, recipient financial institution system **108** also communicates with registration interface **112** and other interfaces according to the configuration of the management portal and the requests received from financial institution clients such as recipient system **110** and sender system **102**. In some examples, recipient financial institution system **108** and sender financial institution system **104** are the same financial institution. In examples where recipient financial institution

system **108** and sender financial institution system **104** are a same financial institution, a same management portal may be used to communication with both registration interface **112** and retrieval interface **114**. In some examples, one financial institution implements one or more management portals.

[0048] Recipient system **110** operates to communicate with recipient financial institution system **108**. In some use cases, recipient system **110** is operated by an individual or business that is receiving the content item and the payment message from another system, such as sender system **102**. Recipient system **110** receives a payment message from a payment network **106** via recipient financial institution system **108**. If the payment message has an associated content item, e.g., includes a GUID, recipient system **110** can request the recipient financial institution system **108** retrieve the content item. Once the recipient financial institution system **108** retrieves the content item from secure content exchange platform **116**, recipient system **110** receives the content item from recipient financial institution system **108**.

[0049] In some embodiments, recipient system **110** views one or both of the payment message and the content item via a portal provided by recipient financial institution system **108**. The portal may be a web-based portal or application, or a mobile application, among other possibilities.

[0050] Registration interface **112** submits content items provided by sender financial institution system **104** with secure content exchange platform **116**. In some embodiments, registration interface **112** also performs checks or verifications of credentials associated with sender financial institution system **104** or sender system **102**. In some embodiments, registration interface **112** is configured, such as through implementation of a service level agreement, to execute registration in a predetermined time period. In one example, the time period for registration is four seconds or less. In some embodiments, registration interface **112** is a registration application programming interface (API), such as a cloud-based API. In some examples, registration interface **112** performs content item checks, such as looking at whether a duplicate of an already registered content item is being submitted. In other examples, such duplicate checks are absent or performed within secure content exchange platform **116**, such as exchange gateway interface **118**.

[0051] Retrieval interface **114** retrieves content items from secure content exchange platform **116** in response to a retrieval request from a verified recipient. In some embodiments, retrieval interface **114** also performs checks or verifications of credentials associated with recipient financial institution system **108** or recipient system **110**. In some implementations, retrieval interface **114** is configured, such as through implementation of a service level agreement, to execute retrieval in a predetermined time period. In one example, the time period for retrieval is two seconds or less. Retrieval interface **114**, in some embodiments, is a retrieval API, such as cloud-based API.

[0052] Other interfaces, not shown, may also be integrated and provide additional functionality to support the content exchange process. Additional examples discussed herein include an access interface and an availability interface, but those of skill in the art will understand that any number of services may be integrated based on desired functionality. Similar to registration interface **112** and retrieval interface **114**, in some embodiments, additional interfaces are implemented as APIs, such as cloud-based APIs.

[0053] An access interface provides security and credential validation for content exchange system **100**. In some embodiments, the access interface enables users to request a token that provides the ability to access interface or perform specific API actions (e.g., register a content item, retrieve a content item, update active status or an expiration date for a content item, etc.) during a particular session. For example, a financial institution, acting as either a sender financial institution or a recipient financial institution, receives a token which is presented to other interfaces, such as registration interface **112** or retrieval interface **114**, as authorization. The token enables a user to access one or more services or submit requests during a session. In some embodiments, the token is configured to expire after a predetermined period of time or following a predetermined number of requests.

[0054] An availability interface provides current status information on a registered document or enables a sender to set a registered document's status as active or inactive, or update its expiration date. In some embodiments, the availability interface is provided by an availability API. In examples, via the availability API, a sender requests to change the status of a content item. A content item's status is active when the content item is presently stored in content database **122** and can be retrieved for recipient system **110** using a GUID associated with the content item. A content item's status may be set to inactive when the content item is removed from content database **122**. An inactive status may also indicate generally that the content item cannot be retrieved, even with proper submission of the associated GUID, although the content item is still maintained in content database **122**.

[0055] A sender, via the availability interface, adjusts an expiration date for the content. In some embodiments, content database **122** provides temporal storage, and a content item stored in content database **122** has an expiration date set by default when the content item is stored in content database **122**. The default may be a predetermined number of days, weeks, month, hours, minutes, etc. that the content item will be maintained in content database **122**. In some implementations, a sender sets an expiration date other than the default during initial registration of a content item or, via the availability interface, adjusts expiration of the content item to be sooner or later than the default. In some embodiments, the default expiration period represents a maximum amount of time a content item is retained within content database **122**, such that a user cannot set the expiration date to be longer than the default date. In one example, the default expiration date is set at 90 days after registration.

[0056] Secure content exchange platform **116** provides secure storage and control of content to be associated with a payment message. In some embodiments, secure content exchange platform **116** includes exchange gateway interface **118**, content exchanger **120**, and content database **122**. In embodiments, secure content exchange platform **116** includes more or fewer components, as functionality is combined or distributed. Because secure content exchange platform **116** maintains content items separate from payment network **106**, content items can be associated with payment messages without burdening the payment network or exposing the content items through conventional exchange means, e.g., email.

[0057] Exchange gateway interface **118** communicates with registration interface **112**, retrieval interface **114**, and content exchanger **120**. Exchange gateway interface **118** regulates traffic and communications into and out of secure content exchange platform **116**. In examples, exchange gateway interface **118** performs conformance checks on incoming content items. Conformation checks may verify a content item's size, type, format, etc.

[0058] Content exchanger **120** generates a GUID and associates the GUID with a received content item. Content exchanger **120** generates the GUID in response to receiving a content item via exchange gateway interface **118**. Content exchanger **120** relays the received content item, in association with the generated GUID, to content database **122** for storage.

[0059] Content database **122** stores registered content items for retrieval. Content database **122** provides a central content repository that provides storage of registered documents. In some embodiments, content database **122** is configured for temporal storage. In an example, content database **122** is configured to automatically delete or otherwise remove any content item older than 90 days.

[0060] FIG. 2 depicts a system-flow diagram illustrating a flow path of a content item **124** through the content exchange system **100** of FIG. 1, according to an example embodiment. Sender system **102** requests to upload and register a content item **124** to content exchange platform **116**. Content item **124** is registered and stored in content database **122**, and a corresponding GUID is sent to sender financial institution system **104**. Content item **124** may be any one of numerous content types, including but not limited to documents, spreadsheets, media files, data files, images, etc.

[0061] FIG. 2 further illustrates how aspects of the present disclosure insulate the content item **124**

from payment network **106**. While communication from sender system **102** to recipient system **110** relays through payment network **106**, content item **124** does not pass through payment network **106**. This arrangement limits the burden on payment network **106**, without exposing the content item **124** through a less secure transmission, such as email. Aspects of the present disclosure also enable direct association between content item **124** and a payment message, rather than the indirect association available with other transmission means, such as email.

[0062] Content item **124** may be any number of content types including, but not limited to, a document, a data file, or an image or other media file. Content item **124** may be submitted or stored as any number of file types based on the configuration of secure content exchange platform **116**. In an example, content exchange platform **116** is configured for a single file type, such as portable document format (PDF) or XML, for simplicity and compactness. In other examples, content exchange platform **116** is configured to accept a range of file formats including, as some nonlimiting examples, compressed files, formatted or plain text files, financial record or financial data transfer formats, graphical information organizer files, scientific data files, presentation files, audio files, video files, encryption files, spreadsheet files, and tabulated data files. In some embodiments, content item **124** is subject to size limitations based on the configuration of secure content exchange platform **116**. In some embodiments, content exchange platform **116** imposes a file size limit of 1 MB, 2 MB, 3 MB, etc. depending on configuration, capacity, and expected turnover. In some embodiments, files in excess of content exchange platform **116** size limits are rejected or compressed.

[0063] As illustrated at **202-210**, FIG. 2 illustrates a registration pathway for content item **124**. Registration of content item **124** allows secure storage of content item **124** and association with a payment message, through an associated GUID, without compromising the security of the payment message or burdening the payment network.

[0064] As illustrated at **202**, sender system **102** submits a content item **124** to sender financial institution system **104** and requests to register the content item **124** with secure content exchange platform **116**. For example, sender system **102** submits a payment message to sender financial institution system **104** with content item **124** associated, with the request to register the content item **124** with secure content exchange platform **116** included through the association between content item **124** and the payment message.

[0065] As illustrated at **204**, sender financial institution system **104** relays the content registration request, with content item **124**, along to secure content exchange platform **116** via registration interface **112**. Registration interface **112** may perform credential validation of sender financial institution system **104**. For example, sender financial institution system **104** provides a key or other identifier establishing it as a known financial institution along with the content item and registration request, to demonstrate the request is being submitted from a trustworthy source. In some embodiments, sender financial institution system **104** first communicates with an access interface and receives a session key or token to be presented to registration interface **112**, to demonstrate sender financial institution's **104** permissions to make use of content exchange platform **116**.

[0066] In some embodiments, content item **124** is encrypted at the relay to registration interface **112**, illustrated at **204**. In some implementations, content item **124** is encrypted by sender financial institution system **104** before being relayed to registration interface **112** using an encrypted communication channel. In other implementations, content item **124** is encrypted by registration interface **112**. Content item **124** may remain encrypted until retrieved, remaining encrypted in memory through exchange gateway interface **118** and content exchanger **120**, and in storage in content database **122**. In some embodiments, content item **124** is encrypted throughout its travel into, through, and out of secure content exchange platform **116**, as illustrated from **204** to **218**.

[0067] As illustrated at **206**, content item **124** is relayed to exchange gateway interface **118** of secure content exchange platform **116** to be registered. Exchange gateway interface **118** may

perform conformance checks on content item **124** to ensure content item **124** is compatible with content exchange platform **116**, such as verifying the content item **124** is a file type and size that is compatible with content database **122**.

[0068] As illustrated at **208**, content exchanger **120** receives content item **124** via exchange gateway interface **118** and registers the content item **124**, generating a corresponding GUID. When content exchanger **120** receives content item **124**, content exchanger generates a new and unique GUID, which content exchanger **120** associates with content item **124** before relaying the content item **124**, with its association to the generated GUID, to content database **122** for storage. As illustrated at **210**, content item **124** is stored in content database **122**.

[0069] Sender system **102** may not be able to register the content item **124** to content exchange platform **116**. In such case sender system **102** is alerted to the registration failure. A sender system **102** may not be able to register the content item **124** if, for example, the operation or security requirements of content exchange system **100** are not satisfied. Some nonlimiting examples of requirements that may result in registration failure are failed credentials by sender system **102** or sender financial institution system **104**, incorrectly formatted metadata on content item **124**, content or document type not being of an accepted type, and content in excess of size limits. In some embodiments, an error message indicating a content item cannot be registered is also generated in response to errors within secure content exchange platform **116**, such as a failure of content exchanger **120** to generate a GUID. The error message is provided to sender financial institution system **104** or, in embodiments, to sender system **102**. In some implementations, the error message provides guidance on the required content and format or specify accepted content types and file size limits or other corrective measures.

[0070] As illustrated at **212-220**, FIG. 2 depicts a retrieval pathway of content item **124**. Content item **124** is retrieved in response to a validated retrieval request. A retrieval request is initiated by recipient system **110**. Recipient system **110** receives a payment message which includes an indication that a content item is associated with the payment message. Recipient system **110** requests to view the content item, such as by selecting a link, button, or menu within or otherwise associated with the payment message display, indicating an associated content item. Recipient financial institution system **108** extracts the associated GUID from the payment message and relays it to retrieval interface **114**.

[0071] Validation of a retrieval request may be performed by retrieval interface **114** or exchange gateway interface **118**, or retrieval interface **114** and exchange gateway interface **118** may perform different aspects of the request validation. In some embodiments, validation of a retrieval request includes checking credentials for recipient financial institution system **108**, checking the GUID itself for proper characteristics, or comparing recipient and recipient financial institution identification data with address metadata associated with content item **124**.

[0072] Retrieval interface **114** may perform credential validation of recipient financial institution system **108**, before relaying the retrieval request to exchange gateway interface **118** of secure content exchange platform **116**. For example, recipient financial institution system **108** provides a key or other identifier establishing it as a known financial institution along with a GUID and retrieval request, to demonstrate the request is being submitted from a trustworthy source. In examples, recipient financial institution system **108** first communicates with an access interface and receive a session key or token to be presented to retrieval interface **114**, to demonstrate recipient financial institution's **108** permissions to make use of content exchange platform **116**.

[0073] Checking the GUID may include checking the formatting and integrity of the GUID, or checking if the GUID is active. Recipient financial institution system **108**, and, in embodiments, recipient system **110**, submit credentials, such as a session token, which are checked for validity, currency, or association with the GUID.

[0074] As illustrated at **212**, content item **124** is retrieved from content database **122** by content exchanger **120**. As illustrated at **214**, content item **124** is relayed to exchange gateway interface **118**

by content exchanger **120**. As illustrated at **216**, content item **124** is related to retrieval interface **114** by exchange gateway interface **118**. As illustrated at **218**, content item **124** is relayed to recipient financial institution system **108** by retrieval interface **114**. As illustrated at **218**, recipient financial institution system **108** receives content item **124**.

[0075] In some embodiments, content item **124** is decrypted when exiting secure content exchange platform **116**, through retrieval interface **114**. In some implementations, recipient financial institution system **108** decrypts content item **124** as it is received, as illustrated at **218**. In other implementation, retrieval interface **114** decrypts content item **124** while relaying content item **124** to recipient financial institution system **108**. As illustrated at **220**, recipient financial institution system **108** presents content item **124** to recipient system **110**.

[0076] FIG. **3** depicts a system-flow diagram illustrating a flow path of a GUID **126** through the content exchange system **100**, according to an example embodiment. While content item **124** is retained in content database **122** and isolated from payment network **106**, GUID **126**, being associated with content item **124**, provides a secure link between a payment message and content item **124**. GUID **126** may be embedded in a payment message, providing a secure link from that payment message to content item **124**, while content item **124** is secured in content database **122**.

[0077] As illustrated at **302-306**, FIG. **3** depicts the generation path for GUID **126**. GUID **126** is generated in response to registration of a content item, such as content item **124**. In some embodiments, GUID **126** is a string of characters, such as alphanumeric characters. The length of the string may be determined according to security considerations and acceptance criteria to be embedded or otherwise include in a payment message. In examples, GUID **126** is between 30 and 60 characters, such as being a 36 character alphanumeric string or a 55 character alphanumeric string.

[0078] As illustrated at **302**, content exchanger **120** generates GUID **126** corresponding to content item **124** once registered. Content exchanger **120** receives content item **124** from exchange gateway interface **118** and, in response, generates a new GUID **126**. GUID **126** is associated with content item **124** and content item **124** is stored in content database **122**. In some embodiments, GUID **126** is associated with content item **124** through metadata or through being stored along with content item **124** as a pair.

[0079] As illustrated at **304**, GUID **126** is relayed to registration interface **112** by exchange gateway interface **118**. As illustrated at **306**, GUID **126** is relayed to sender financial institution system **104** by registration interface **112**.

[0080] As illustrated at **308-310**, FIG. **3** depicts transmission of GUID **126**, as part of a payment message, through payment network **106**. GUID **126** provides a secure link between a payment message and a content item, such as content item **124**, enabling secure association of referential content items within the existing infrastructure of a payment network.

[0081] As illustrated at **308**, sender financial institution system **104** includes GUID **126** in a payment message routed to recipient financial institution system **108** via payment network **106**. In some embodiments, GUID **126** is embedded in the payment message or inserted in an electronic address field. In embodiments, the electronic address field accepts one or more GUIDs. In embodiments, the payment message and included GUID are addressed to a single recipient, or multiple recipients. The payment message may be any message transmitted over a secure payment network **106**, and may not particularly be a message transferring or requesting funds.

[0082] As illustrated at **310**, recipient financial institution system **108** receives the payment message and extracts GUID **126**. Recipient financial institution system **108** presents the payment message to recipient system **110**. Recipient system **110** may request to view the associated content item and, in response to recipient's **110** request, recipient financial institution system **108** extracts GUID **126** from the payment message.

[0083] As illustrated at **312-316**, FIG. **3** depicts a retrieval path for a content item, such as content item **124**. Retrieval of a content item is executed by recipient financial institution system **108** in

response to a request to review a content item by recipient system **110**. Recipient financial institution system **108** uses GUID **126**, received within a payment message via payment network **106**, to identify the content item to be retrieved.

[0084] As illustrated at **312**, in response to a retrieval request from recipient system **110**, recipient financial institution system **108** submits GUID **126** to retrieval interface **114**. As illustrated at **314**, retrieval interface **114** relays the retrieval request to exchange gateway interface **118** of secure content exchange platform **116**. As illustrated at **316**, content exchanger **120** receives GUID **126** and retrieves corresponding content item **124** from content database **122**.

[0085] FIG. **4** illustrates a flow chart of a method **400** for content registration of a content item, according to an example embodiment. Registration of a content item is the process by which a content item, submitted by a sender, is associated and stored with a GUID within the secure content exchange platform. Example exceptions which may prevent a content item from being registered include failed credentials, incorrectly formatted metadata, or an unaccepted content or file type.

[0086] At step **402**, a sender requests to register a content item. The sender generally directs the request to register the content item to their sender financial institution. In some embodiments, the registration request is included with a request to transmit a payment message. In embodiments, the process or portal used by the sender to submit the content item and registration request includes credentialling or verification of the sender or the content item. While the present example presents an example registration of a single content item, in embodiments multiple content items are submitted and registered together.

[0087] At step **404**, the sender financial institution relays the registration request to the content exchange platform. The registration request is relayed to the content exchange platform via a registration interface, such as a registration API. In embodiments, relaying the registration request to the content exchange platform includes credentialling or validation of the sender financial institution or the content item. Credentialling of the sender financial institution may be performed by a dedicated access interface, such as an access API.

[0088] The access API is configured to provide a participating financial institution an access token to interact with content exchange services or APIs. In examples, access tokens are active only for a given session. In some embodiments, the token is configured to maintain a session for a predetermined period of time, or to support a specific request or number of requests.

[0089] At step **406**, a determination is made whether or not the platform is available. If no at step **406**, and the platform is not available, a retry limit is checked at step **408**. If no at step **408**, and the retry limit has not been reached, a retry is performed and the availability of the platform is checked again at step **406**. If yes at step **408**, and the retry limit has been reached, the platform is determined to be unavailable, at step **410**. The sender financial institution will receive a notification that the content exchange platform is unavailable and may initiate troubleshooting.

[0090] If yes at step **406**, and the platform is available, the sender financial institution's credentials are validated, at step **412**. If no at step **412**, and the sender financial institution's credentials are not validated, the sender financial institution is notified at step **410**. Failed credentials are one potential cause of registration failure. To ensure security of stored content items, a financial institution must establish their credentials before relaying content items into, or receiving content items out of, the secure content exchange platform. The sender may be expected to reinitiate the registration request. The sender financial institution may be expected to inform the sender that the registration request may need to be reinitiated. If additional technical support is necessary, such as to update credentials, in some embodiments, the sender financial institution is the party to communicate with technical support associated with the content exchange platform.

[0091] If yes at step **412**, and the sender financial institution's credentials are validated, the content item itself is checked for conformance, at step **414**. Conformance checks may verify, for example, a content item's metadata formatting, the content type, and the content item size. If no at step **414**, and the content item fails the conformance checks, the sender may be notified, at step **416**. In some

embodiments, conformance checks are applied to content items being submitted to the content exchange platform to ensure received items comply with the necessary parameters to allow registration. Conformance checks are performed, by way of example, by an exchange gateway interface of the secure content exchange platform, such as exchange gateway interface **118** of FIG. **1**.

[0092] Examples of conformance failures that may preclude a content item from being registered with the content exchange platform include content types that the platform is not configured to accept, file types the platform is not configured to accept, and content items with incorrectly formatted metadata. For example, for a content item that cannot be registered due to its metadata being formatted incorrectly, the sender may be expected to review metadata associated with the registration request to ensure it is correct or update the content type to an accepted format. In some embodiments, the sender financial institution is expected to add or adjust business rules restricting the sender from inputting or uploading incorrect metadata or content types that are not accepted. The sender financial institution, in some implementations, is also expected to inform the sender that the content type is not accepted and provide additional guidance on accepted formats (e.g., PDF, XML, etc.).

[0093] Another example of conformance failure that may prevent registration is content items or files that exceed size limits of the content exchange platform. In some embodiments, the sender is expected to update the file size of the content to an acceptable range. In an example, the acceptable range is less than 3 MB. The sender financial institution may be expected to inform the sender that their content cannot be registers because it exceeds size limits and to implement or update business rules to restrict senders from uploading content files in excess of content size limits.

[0094] If yes at step **414**, and the content item relays the conformance checks, the content item is registered with the content exchange platform, at step **418**. At step **420**, a GUID is generated. At step **422**, the content item is stored in the content database. At step **424**, the GUID is relayed to the sender FI.

[0095] At step **426**, a check is performed to verify the GUID has been received by the sender FI. If no at step **426**, and the GUID was not received, the sender FI is notified, at step **410**. Another example cause of registration failure is if GUID generation is unsuccessful. In this example, the registration process gets as far as the content item being received by the content exchange platform, but an error prevents the GUID from generating correctly, proper association of the content item and the GUID, or proper storage of the content item in association with the GUID. In some embodiments, the sender is expected to review the content item and the registration request to ensure there are no errors. The sender financial institution informs the sender the content item failed to upload to the content exchange platform and may coordinate with technical services for the content exchange platform for troubleshooting.

[0096] If yes at step **426**, and the sender financial institution successfully receives the GUID, the GUID is inserted into the payment message, at step **428**. The sender financial institution includes the GUID in a payment message routed to the recipient financial institution via a payment network. In some embodiments, the payment message is an Extensible Markup Language (XML) message such as International Organization for Standardization (ISO) 20022 defined messages (e.g., pain.013 message, pacs.008 message, etc.). The payment message may be requesting payment, e.g., an RfP, or initiation of a payment, e.g., a payment credit message.

[0097] The sender financial institution then transmits a payment message including the GUID to the appropriate recipient financial institution via a payment network. In some embodiments, the payment network is a real-time payment network. The recipient financial institution receives the payment message and presents the payment message to the recipient. The recipient may verify the details of the payment message, such as the sender, a bill summary, an amount due, a due date, an account number, etc.

[0098] Method **400** presents an example where the sender makes a single request to the sender

financial institution to both prepare the payment message and register the content item. This may be preferred in some instances as it minimizes relaying of the GUID and provides greater streamlining from the client's perspective. However, other embodiments methods are contemplated where, for example, a sender first requests registration of the content item and, in a separate request, requests a payment message with the GUID associated with the content item included. This may be preferred in some instances as providing greater flexibility to the sender to associate a particular content item with more than one payment message.

[0099] FIG. 5 illustrates a flow chart of a method **500** for content retrieval, according to an example embodiment. A recipient financial institution communicates with a secure content exchange platform to retrieve a content item associated with a GUID extracted from a payment message. The recipient financial institution communicates with the content exchange platform on behalf of the recipient. The recipient financial institution provides communications to the recipient on behalf of the secure content exchange platform, for example if an error occurs or the content item is not able to be retrieved for another reason. A recipient may be unable to retrieve a content item if, for example, there is a problem with the recipient's or the recipient financial institution's credentials or if the submitted GUID is not active. In some embodiments, a content item can be retrieved any number of times as long as the GUID remains active and the temporal storage of the content item within the content database has not expired. In embodiments, registration of a content item indicates a predetermined number of times a content item is able to be retrieved.

[0100] At step **502**, a recipient requests that a content item, associated with a payment message received by the recipient, be retrieved. For example, the recipient clicks or otherwise selects a link, button, or menu appearing on or with the payment message that indicates additional content is associated with the payment message. At step **504**, a recipient financial institution, supporting the recipient, relays the recipient's request to the secure content exchange platform. The message may be relayed via a retrieval interface.

[0101] At step **506**, a determination is made whether the secure content exchange platform is available. If no at step **506**, and the content exchange platform is not available, a determination is made if a retry limit has been reached, at step **508**. If no at step **508**, and the retry limit has not been reached, another attempt to reach the content exchange platform is made, at step **506**. If yes at step **508**, and the retry limit has been reached, the recipient financial institution is notified, at step **510**. If the content exchange platform is down or otherwise unavailable for some time, the recipient financial institution may wait a period of time and try again or reach out to technical support.

[0102] If yes at step **506**, and the content exchange platform is available, the recipient financial institution's credentials are validated at step **512**. If no at step **512**, and the recipient financial institution's credentials cannot be validated, the recipient financial institution is notified, at step **510**. If yes at step **512**, and the recipient financial institution's credentials are validated, a determination is made whether the GUID is active, at step **514**.

[0103] If no at step **514**, and the GUID is not active, the recipient financial institution is notified, at step **510**. A GUID may be deactivated if, for example, the sender removes the content item from the content exchange platform. In some instances, a GUID is deactivated by a sender updating a content item on the content exchange platform. If yes at step **514**, and the GUID is active, the GUID is processed and matched to a content item, at step **516**.

[0104] The content item matched to the GUID, in some embodiments, includes metadata defined by the sender identifying the intended recipient financial institution system or recipient system. The recipient financial institution system from which the retrieval request is received is compared to the recipient financial institution system identified in the metadata, and a determination is made whether the requesting recipient financial institution system matched the metadata identified recipient financial institution system, at step **517**.

[0105] If no at step **517**, and the determination is made that the requesting recipient financial institution does not match the metadata identified recipient financial institution, the requesting

recipient financial institution is notified, at step **510**. If yes at step **517**, and the determination is made that the requesting recipient financial institution does match the metadata identified recipient financial institution, the content item associated with the GUID is fetched, at step **518**. At step **520**, the content item is relayed to the recipient financial institution. At step **522**, the content item is presented to the recipient.

[0106] FIG. **6** illustrates a flow chart of a method **600** for updating a registered content item, according to an example embodiment. A sender requests to update or replace a content item associated with a particular GUID. The original GUID may be maintained and linked to a new or updated content item. In some embodiments, the original content item is no longer be available to the recipient following the change.

[0107] At step **602**, a sender requests to update or replace a registered content item. In embodiments, the sender requests to remove a registered content item from the content exchange platform without replacing it. At step **604**, the sender financial institution relays the update request to the content exchange platform.

[0108] At step **606**, the sender financial institutions credentials are validated. If no at step **606**, and the sender financial institutions credentials are not validated, the sender financial institution is notified, at step **608**. If yes at step **606**, and the sender financial institutions credentials are validated, the sender's permissions are verified, at step **610**. Verifying the sender's permissions may also include performing conformance checks on the new or updated content item or comparing a sender's identification with metadata identifying the sender associated with the submission of the original content item.

[0109] If no at step **610**, and the sender's permissions are not verified, the sender is notified, at step **612**. If yes at step **610**, and the sender's permission are verified, the new or updated content item is associated with the original GUID, at step **614**. At step **616**, the new or updated content item is stored. In some embodiments, the original content item is deleted or otherwise removed from the content database.

[0110] At step **618**, confirmation of successful updating or replacement of the content item is sent to the sender financial institution to relay to the sender. If, instead, the content item cannot be updated or replaced due to an error, an update failure notification may instead be sent.

[0111] FIG. **7** illustrates an example communication sequence **700** between a financial institution infrastructure **702** and secure content exchange platform **116**. The financial institution infrastructure **702** may be a sender financial institution, a recipient financial institution, or both depending on the requests received from clients.

[0112] The financial institution operating the financial institution infrastructure **702** initiates a session by requesting a token **704**, such as through an access interface. Once the financial institution infrastructure **702** receives the token **706**, the session is opened and remains open as long as the token is active. In some embodiments, the token is configured to remain active for a predetermined period of time, to support a predetermined number of requests, to support a particular request, until an error is thrown, or to be deactivated by request from the financial institution infrastructure **702**.

[0113] Once the session is open, financial institution infrastructure **702** may execute one or more process flow in communication with secure content exchange platform **116**. For example, a registration request **708** is sent on behalf of a sender, and a registration notification and GUID **710** received in return. In some embodiments, the registration request includes a header with one or more credentials or other metadata elements. For example, a registration request header includes a session token **712**, an identifier **714** associated with the financial institution infrastructure **702**, and, in embodiments, an identifier **716** associated with the sender. In examples, the registration request header includes additional metadata, such as content type, format, and size. Additional metadata, such as an address identifier associated with a recipient or an address identifier associated with a recipient financial institution is included in some embodiments.

[0114] Financial institution infrastructure **702** may be configured to communicate with both registration and retrieval interfaces. In some embodiments, the same financial institution infrastructure **702** submits, on behalf of a recipient, a retrieval request **718** and receives a content item **720** in return. In an example, the content item registered with process flows including registration request **708** and GUID **710** are the same content item retrieved with process flows including retrieval request **718** and content item **720**. Retrieval request **718** includes a header with one or more credentials or other metadata elements, such as session token **712**, an identifier **714** associated with the financial institution infrastructure **702**, and, in embodiments, an identifier **722** associated with the recipient. In this example, session token **712** is the same because it is a same session from view of financial institution infrastructure **702** and a same identifier **714** because it is the same financial institution infrastructure **702**, but in embodiments different session tokens and financial institution infrastructures, and associated identifiers, are involved.

[0115] Communication sequence **700** is applicable, in embodiments, to registration, retrieval, or update processes, and include communication with other services such as access interfaces and availability interfaces. For example, similar process flows are used by a financial institution using an availability interface to determine a status or history of a registered content item.

[0116] FIG. **8** illustrates an example block diagram of a computing system **800**. Computing system **800** may be virtual or physical. One or more aspects of the computing system **800** can be used to implement the systems and methods described above in conjunction with FIGS. **1-7**, including sender system **102**, sender financial institution system **104**, payment network **106** infrastructure, recipient financial institution system **108**, recipient system **110**, registration interface **112**, retrieval interface **114**, and secure content exchange platform **116**, including exchange gateway interface **118** and content exchanger **120**.

[0117] In an example, the computing system **800** can include a computing environment **802**. The computing environment **802** can be a physical computing environment, a virtualized computing environment, or a combination thereof. The computing environment **802** can include memory **804**, a communication medium **812**, one or more processing units **814**, a network interface **816**, and an external component interface **818**.

[0118] The memory **804** can include a computer readable storage medium. The computer storage medium can be a device or article of manufacture that stores data and/or computer-executable instructions. The memory **804** can include volatile and nonvolatile, transitory and non-transitory, removable and non-removable devices or articles of manufacture implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. By way of example, and not limitation, computer storage media may include dynamic random access memory (DRAM), double data rate synchronous dynamic random access memory (DDR SDRAM), reduced latency DRAM, DDR2 SDRAM, DDR3 SDRAM, solid state memory, read-only memory (ROM), electrically-erasable programmable ROM, optical discs (e.g., CD-ROMs, DVDs, etc.), magnetic disks (e.g., hard disks, floppy disks, etc.), magnetic tapes, and other types of devices and/or articles of manufacture that store data.

[0119] The memory **804** can store various types of data and software. For example, as illustrated, the memory **804** includes software application instructions **806**, one or more databases **808**, as well as other data **810**.

[0120] In addition, the software application instructions **806** can be included in non-transitory computer-readable medium having stored thereon one or more sequences of instructions, which when executed by one or more processors, cause the one or more processors to perform methods or implement systems described above in conjunction with FIGS. **1-7**.

[0121] The communication medium **812** can facilitate communication among the components of the computing environment **802**. In an example, the communication medium **812** can facilitate communication among the memory **804**, the one or more processing units **814**, the network interface **816**, and the external component interface **818**. The communication medium **812** can be

implemented in a variety of ways, including but not limited to a peripheral component interconnect (PCI) bus, a PCI express bus accelerated graphics port (AGP) bus, a serial Advanced Technology Attachment (ATA) interconnect, a parallel ATA interconnect, a Fiber Channel interconnect, a USB bus, a Small Computing system interface (SCSI) interface, or another type of communications medium.

[0122] The one or more processing units **814** can include physical or virtual units that selectively execute software instructions, such as the software application instructions **806**. In an example, the one or more processing units **814** can be physical products comprising one or more integrated circuits. The one or more processing units **814** can be implemented as one or more processing cores. In another example, one or more processing units **814** are implemented as one or more separate microprocessors. In yet another example embodiment, the one or more processing units **814** can include an application-specific integrated circuit (ASIC) that provides specific functionality. In yet another example, the one or more processing units **814** provide specific functionality by using an ASIC and by executing computer-executable instructions.

[0123] The network interface **816** enables the computing environment **802** to send and receive data from a communication network. The network interface **816** can be implemented as an Ethernet interface, a token-ring network interface, a fiber optic network interface, a wireless network interface (e.g., Wi-Fi), or another type of network interface.

[0124] The external component interface **818** enables the computing environment **802** to communicate with external devices. For example, the external component interface **818** can be a USB interface, Thunderbolt interface, a Lightning interface, a serial port interface, a parallel port interface, a PS/2 interface, or another type of interface that enables the computing environment **802** to communicate with external devices. In various embodiments, the external component interface **818** enables the computing environment **802** to communicate with various external components, such as external storage devices, input devices, speakers, modems, media player docks, other computing devices, scanners, digital cameras, and fingerprint readers.

[0125] Although illustrated as being components of a single computing environment **802**, the components of the computing environment **802** can be spread across multiple computing environments **802**. For example, one or more of instructions or data stored on the memory **804** may be stored partially or entirely in a separate computing environment **802** that is accessed over a network.

[0126] Depending on the size and scale of the computing environment **802**, it may be advantageous to include one or more load balancers to balance traffic across multiple physical or virtual machine nodes.

[0127] Aspects of the computing system **800** and the computing environment **802** can be protected using a robust security model. In an example, users may be made to sign into the system using a directory service, such as ACTIVE DIRECTORY by MICROSOFT CORPORATION of Redmond, Washington. Connection and credential information can be externalized from jobs using an application programming interface. Credentials can be stored in an encrypted repository in a secured operational data store database space. Privileges can be assigned based on a collaboration team and mapped to a Lightweight Directory Access Protocol (LDAP) Group membership. A self-service security model can be used to allow owners to assign other's permissions on their objects (e.g., actions).

[0128] Each node may be configured to be capable of running the entirety of the computing system **800**, such that portal can run and schedule jobs and serve the portal user interface as long as a single node remains functional. The computing environment **802** may include monitoring technology to determine when a node is not functioning so an appropriate action can be taken.

[0129] In addition, it should be understood that FIGS. 1-8 are presented for example purposes only. The architecture of the example embodiments presented herein is sufficiently flexible and configurable, such that it may be utilized (and navigated) in ways other than that shown in the

accompanying figures.

[0130] Having described the preferred aspects and implementations of the present disclosure, modifications and equivalents of the disclosed concepts may readily occur to one skilled in the art. However, it is intended that such modifications and equivalents be included within the scope of the claims which are appended hereto.

Claims

1-22. (canceled)

23. A method for secure content exchange, comprising: receiving, from a sending financial institution system, a request for a session token; generating the session token with a predetermined active period; validating the session token for subsequent requests from the sending financial institution system; receiving, within the predetermined active period, a registration request for a content item; verifying credentials in a registration request header, the credentials comprising: the session token; a financial institution identifier; and a sender identifier; encrypting the content item; storing the encrypted content item in a content database; generating a globally unique identifier (GUID) associated with the encrypted content item; and returning the GUID to the sending financial institution system.

24. The method of claim 23, further comprising: receiving an availability request for the content item; determining a status of the content item, wherein the status comprises: an active status indicating the content item is available for retrieval; an inactive status indicating the content item is not available for retrieval; or an expired status indicating the content item has been deleted from the content database; validating the session token associated with the availability request; and returning the status of the content item.

25. The method of claim 23, further comprising: receiving a retrieval request including the GUID from a recipient financial institution system; verifying retrieval credentials through an access interface, wherein the retrieval credentials comprise a retrieval token providing authorization to perform specific retrieval actions during a particular session; validating that the recipient financial institution system matches recipient metadata associated with the encrypted content item; and determining that both: the retrieval credentials are successfully verified through the access interface; and the recipient financial institution system is successfully validated as matching the recipient metadata; and returning the encrypted content item upon the determination that both the verification and validation are successful.

26. The method of claim 25, wherein verifying the retrieval credentials comprises: receiving a token request from the recipient financial institution system; generating the retrieval token with a predetermined configuration comprising at least one of: a specified time period for token validity; a maximum number of retrieval requests; or authorization for a specific retrieval action; and validating the retrieval token before processing the retrieval request.

27. The method of claim 23, wherein storing the encrypted content item comprises: validating metadata associated with the content item, wherein the metadata comprises: a sender financial institution identifier; a recipient financial institution identifier; content type information; content size information; and content format specifications; and storing the validated metadata with the encrypted content item in the content database.

28. The method of claim 23, wherein the predetermined active period comprises at least one of: a specified time duration; a maximum number of requests; or completion of a specific request.

29. The method of claim 23, further comprising: executing one or more process flows in communication with the secure content exchange platform within the predetermined active period using the session token.

30. The method of claim 23, wherein the content item comprises at least one of: a document; a data file; an image; or a media file.

- 31.** The method of claim 23, further comprising: validating file size of the content item against predetermined size limits prior to storage, wherein: the predetermined size limits are configured according to content exchange platform capacity; content items exceeding size limits are rejected or compressed; and size limit validation occurs during conformance checks by the exchange gateway interface.
- 32.** The method of claim 23, further comprising: maintaining the content item in an encrypted state: during initial transmission from the sending financial institution system; throughout storage in the content database; during retrieval transmission to a recipient financial institution system; and until decryption by the recipient financial institution system or retrieval interface.
- 33.** A system for secure content exchange, comprising: a content database configured to: store a content item; associate an expiration date with the stored content item; and automatically delete the content item when the expiration date is reached; a content exchanger configured to: generate a globally unique identifier (GUID) associated with the content item; encrypt the content item prior to storage in the content database; and an exchange gateway interface configured to: receive a registration request including the content item; validate a default expiration period for the content item; submit the content item to the content exchanger; receive the GUID associated with the content item; return the GUID as a response to the registration request; and maintain the content item in an encrypted state during transmission.
- 34.** The system of claim 33, wherein the content database is configured to: store the content item with a configurable expiration date; receive, via an availability interface, a request to modify the expiration date for the content item; update the expiration date in response to the request; and automatically delete the content item when the expiration date is reached.
- 35.** The system of claim 33, wherein the exchange gateway interface is further configured to: receive a request to modify the expiration date; and update the expiration date in the content database.
- 36.** The system of claim 33, wherein the exchange gateway interface is further configured to perform conformance checks on the content item to verify compatibility with the content database.
- 37.** The system of claim 36, wherein the conformance checks comprise verifying at least one of: file type compatibility; file size limits; or metadata formatting.
- 38.** The system of claim 33, wherein the content exchanger is further configured to: receive a request to replace the content item with a second content item; encrypt the second content item; store the encrypted second content item in the content database; and associate the GUID with the second content item.
- 39.** The system of claim 33, wherein the exchange gateway interface is further configured to: validate credentials of a recipient financial institution system prior to processing a retrieval request.
- 40.** The system of claim 33, wherein the content database is further configured to: maintain an audit trail of access attempts to the stored content item.
- 41.** The system of claim 33, wherein the exchange gateway interface is further configured to: verify the registration request originates from an authorized financial institution system by: receiving an access token request from the financial institution system; validating the financial institution system's credentials through an access interface; generating a session-specific access token; and verifying the session-specific access token before processing the registration request.
- 42.** The system of claim 33, wherein the content exchanger is further configured to: verify successful encryption of the content item by: confirming encryption before relay to the content database; validating encrypted state maintenance during storage; verifying encryption status during retrieval operations; and maintaining encryption through the exchange gateway interface transmission.
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